

Permit to Work and Safety-Critical Task Standard

VIN-SOP-MNT-412

PLANT MAINTENANCE DIVISION

STANDARD OPERATING PROCEDURE: PERMIT-TO-WORK SYSTEM

1. PURPOSE AND SCOPE

1.1 This standard defines the mandatory governance, authorization hierarchy, and procedural steps for executing high-risk maintenance and engineering tasks within Vindhya Precision Works (VPW).

1.2 Compliance with this procedure is mandatory for all direct employees, contractors, and visiting specialists operating within VPW manufacturing blocks, utility yards, and sub-stations.

2. DEFINITIONS AND TASK CLASSES REQUIRING A PERMIT

No high-risk work shall commence without an active, signed Permit-to-Work (PTW) issued in accordance with Section 4. The following five task classes require a dedicated PTW:

2.1 Hot Work (Class HW): Any operation involving open flames, electric arc welding, oxy-fuel cutting, grinding, or generation of sufficient spark energy to ignite combustible atmospheres.

2.2 Confined Space Entry (Class CS): Entry into any enclosure that has limited openings for entry and exit, is not designed for continuous employee occupancy, and has potential for accumulated hazardous atmospheres, oxygen deficiency, or engulfment.

2.3 Work at Height (Class WAH): Any work activity where a person could fall a vertical distance of 1.8 meters or greater from an unprotected edge, scaffold, ladder, or elevated platform.

2.4 High-Voltage Electrical Work (Class HV): Any maintenance, testing, or modification on electrical systems operating at voltages exceeding 1,000V AC or 1,500V DC.

2.5 Pressure System Breaking (Class PSB): The mechanical isolation, blinding, opening, or breaking of containment on any piping, vessel, or heat exchanger operating above 6.0 bar gauge or containing hazardous process media regardless of pressure.

3. PERMIT VALIDITY PERIOD AND EXTENSIONS

3.1 Standard PTW validity shall not exceed twelve (12) consecutive hours from the time of issue.

3.2 Hot Work permits issued inside Class-1 hazardous process areas are restricted to a maximum validity of eight (8) hours.

3.3 A permit may be extended precisely once, for a maximum duration of four (4) hours, provided:

- a) The issuing authority inspects the worksite and verifies site conditions remain unchanged.
- b) Gas testing is repeated and recorded within thirty (30) minutes prior to extension authorization.
- c) Shift handover logs are countersigned by both the outgoing and incoming Permit Acceptors.

4. NAMED ROLES: ISSUANCE AND ACCEPTANCE

Permits must be authorized through the dual-key sign-off of two distinct named roles:

4.1 Permit Issuer (PI): The authorized individual holding statutory delegation from the VPW Plant Engineering Manager. Designated PIs for Vindhya Precision Works are:

- a) Shri V. K. Ranganathan (Senior Utilities Lead - Electrical/Pressure)
- b) Smita Deshmukh (Mechanical Maintenance Supervisor - Hot Work/Confined Space)
- c) Rajeshwar Rao (Safety & Compliance Officer - All Classes)

4.2 Permit Acceptor (PA): The competent contractor supervisor or VPW lead technician in direct physical control of the task crew. PAs must hold current VPW-internal certification for the specific task class and possess a valid Level-3 Safety Passport.

5. GAS-TESTING REQUIREMENTS AND RE-TEST INTERVALS

5.1 Mandatory atmospheric testing using a calibrated multi-gas monitor (detecting Oxygen, Lower Explosive Limit [LEL], Carbon Monoxide, and Hydrogen Sulphide) must be performed by a certified tester prior to permit issuance for Class CS, Class HW, and Class PSB.

5.2 Safe entry thresholds are strictly defined as:

- a) Oxygen: 19.5% to 23.5% by volume.
- b) LEL: Less than 1.0%.
- c) Carbon Monoxide: Below 25 ppm.
- d) Hydrogen Sulphide: Below 5 ppm.

5.3 Re-test Intervals:

- a) For Confined Space Entry (Class CS), continuous monitoring is mandatory, with manual verification logging every two (2) hours.
- b) For Hot Work (Class HW) and Pressure Breaking (Class PSB) in open or semi-enclosed spaces, re-testing must occur at intervals not exceeding four (4) hours.
- c) Any work stoppage exceeding thirty (30) minutes requires a full baseline re-test before work resumes.

6. CONDITIONS FOR AUTOMATIC PERMIT VOIDING

A PTW is rendered immediately null and void upon the occurrence of any of the following triggers, requiring full re-application and authorization:

- 6.1 Sounding of any plant-wide or unit-specific emergency evacuation alarm.
- 6.2 Unplanned process upset, chemical release, or pressure spike within fifty (50) meters of the worksite.
- 6.3 Gas detector reading exceeding the safe thresholds defined in Section 5.2.
- 6.4 Loss of required site safety infrastructure (e.g., local fire watch leaves station, extraction ventilation failure, or loss of dedicated lighting).
- 6.5 Deviation from the isolation methodology or PPE schedule specified on the permit face.
- 6.6 Expiration of the valid time window stipulated in Section 3.

7. AUTOMATED ADVISORY SYSTEM RESTRICTIONS

7.1 The Vindhya Maintenance Management System (VMMS) automated advisory module is strictly prohibited from generating, validating, or issuing automated recommendations for the following categories without explicit human authorization from the Plant Engineering Manager or Safety & Compliance Officer:

- a) Isolation Bypass Authorization: Recommending the removal, override, or bypassing of Safety Instrumented Systems (SIS), interlocks, or hardware-based energy isolation locks.
- b) Atmospheric Safety Override: Recommending entry into a confined space or initiation of hot work when gas test parameters fall outside the limits specified in Section 5.2.
- c) Pressure System De-isolation: Recommending the pressurization, re-commissioning, or removal of blinds on pressure systems following Class PSB execution.
- d) High-Voltage Earthing Removal: Recommending the removal of temporary earth leads or clearance for re-energization on Class HV electrical assets.

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